Language-Aware Hard Negative Mining for Improved Vision-Language Representation Learning

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Motivation

Why does hard negative mining matter?

Problem with standard contrastive learning:

- CLIP-style training pushes matched (image, text) pairs together and unmatched pairs apart.
- Easy negatives (very different images/captions) provide little gradient signal.
- Hard negatives — pairs that are similar but not matched — are crucial for learning fine-grained distinctions.

The LLaVE insight (our starting point):

- Dynamically reweight negatives by their embedding-space similarity (hardness score).
- Removes the need for separate offline mining stages.
- BUT: hardness computed purely from the model's own embeddings has a flaw...

⚠ *Two captions saying the same thing differently (e.g. "A dog runs" vs "A canine sprints") get treated as hard negatives — wasting model capacity.*

Our Approach

Language-Aware Hardness Score

Key Idea: inject an external linguistic prior (Sentence-BERT)

$$H(i,j) = \alpha \cdot sim_embed(i,j) + \beta \cdot sim_ling(i,j)$$

$$w(i,j) = \exp(H(i,j))$$

sim_embed

- Cosine similarity in the model's embedding space
- Scaled by logit temperature $\tau = 50$
- Updates each training step as model learns

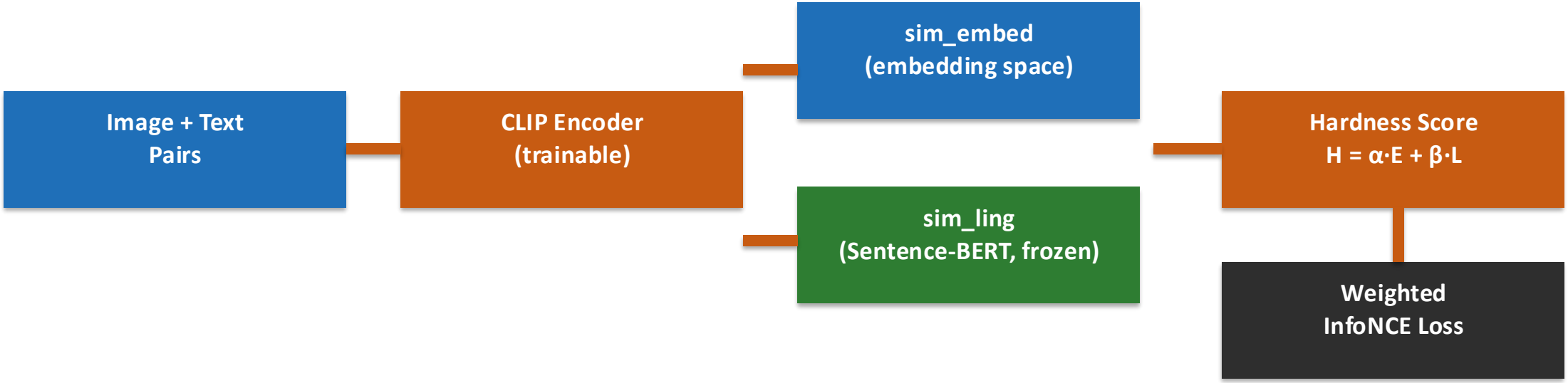
sim_ling (NEW)

- Cosine similarity from frozen Sentence-BERT
- Pre-computed once — zero overhead at train time
- Catches semantically equivalent captions

$\beta = 0 \rightarrow$ standard LLaVE baseline | $\alpha=5, \beta=5 \rightarrow$ our best (Balanced) configuration

Ablation: $(\alpha,\beta) \in \{(9,0), (9,1), (5,5), (9,5), (1,9)\}$ — equal weighting wins

Method Overview



Pre-compute step (simularity.py):
All captions → Sentence-BERT → .pt embeddings saved to disk

At training time: batch indices → O(1) lookup of pre-computed SBERT embeddings. Zero model parameters added.

Implementation Details

Making it work in BFloat16

Two engineering fixes were needed for stable training:

Fix 1

FP32 Upcast

- BFloat16 rounds small values to 0 — loss collapses to 0/0
- Cast sim_embed and sim_ling to float32 before any exponential
- Prevents NaN gradients during training

Fix 2

Log-Sum-Exp Trick

- Naively computing exp(sim) creates huge intermediate values
- Subtract row maximum before exponentiation — numerically identical but stable
- Entire loss stays in log-space; no intermediate 1.0/1.0 rounding

Pre-computation: simularity.py encodes all captions with frozen Sentence-BERT once before training.
Result: (M × 384) tensor saved to disk. Training: batch indices → O(1) lookup. No GPU overhead.

Experimental Setup

Dataset

- Flickr30K — 1,000 test images (Karpathy split)
- 1 caption per image in evaluation
- 1,000-way retrieval: both I→T and T→I

Models

- Backbone: LLaVE-2B (LlavaQwenForCausalLM + SigLIP-so400m-patch14-384)
- Linguistic prior: frozen Sentence-BERT all-MiniLM-L6-v2
- AdamW lr=2e-4, batch 4 × accum 8 = eff. 32, 3 epochs, DeepSpeed ZeRO-2

Hyperparameters Ablated

- (α=9, β=0) baseline | (α=9, β=1) language-heavy
- (α=5, β=5) balanced ★ | (α=9, β=5) high αβ
- (α=1, β=9) embed-heavy — 5 configs total

Evaluation Metrics

- Recall@1 — correct item in top 1
- Recall@5 — correct item in top 5
- Recall@10 — correct item in top 10
- Median Rank (MedR) — lower is better
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- Reported for both directions:
- Image → Text (I→T)
- Text → Image (T→I)

Results on Flickr30K

T→I retrieval — 1K images, 5K captions (evaluate_retrieval.py)

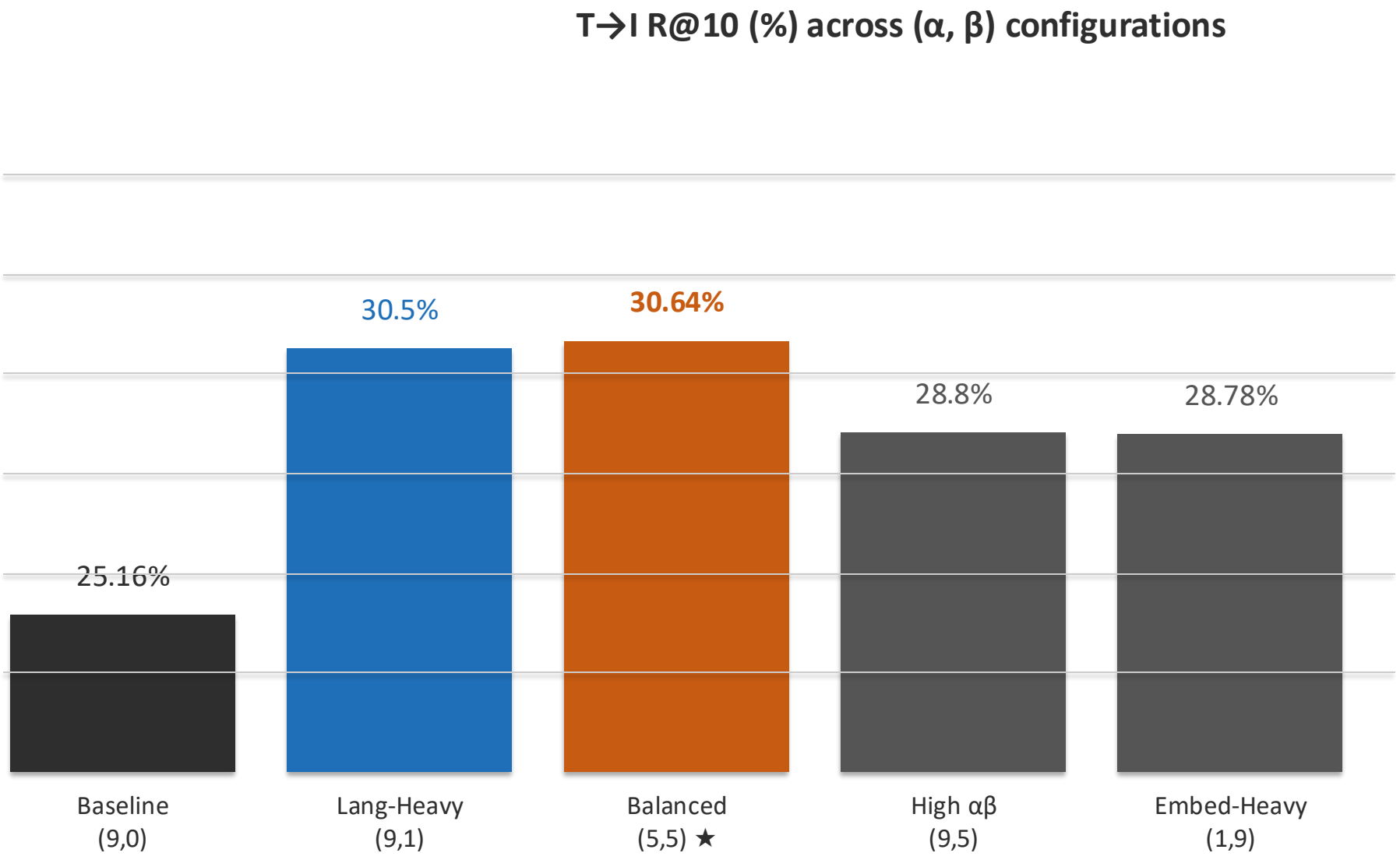
Configuration (α, β)	R@1	R@5	R@10	MedR
Baseline (9, 0)	6.70	17.34	25.16	47
Language-Heavy (9, 1)	8.64	22.00	30.50	33
Balanced (5, 5) ★ BEST	**8.70**	**22.26**	**30.64**	**33**
High αβ (9, 5)	8.04	21.00	28.80	37
Embed-Heavy (1, 9)	7.86	20.84	28.78	36
vs Baseline ↑	*+2.0 pp*	*+4.92 pp*	*+5.48 pp (+21.8% rel)*	*47→33*

Key Takeaways:

- Balanced (α=5, β=5) is best — equal weighting of embedding and linguistic signals.
- R@10 improves 25.16% → 30.64% (+21.8% relative); MedR drops from 47 to 33.
- High αβ (9,5) underperforms Balanced (5,5) despite same β — gradient conflict.

Ablation Study

Effect of (α, β) balance — T→I R@10 on Flickr30K



Observations:

- Adding β=1 to baseline already closes most of the gap.
- Balanced (5,5) edges out (9,1) — equal weighting is optimal.
- High αβ (9,5) drops vs. Balanced: gradient conflict.
- Embed-Heavy (1,9): over-weighting text hurts.
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- → Equal α=β=5 provides the most stable gradient signal.

Conclusion & Future Work

What we did:

- Extended LLaVE's hardness-weighted contrastive loss with a frozen Sentence-BERT linguistic prior.
- Combined embedding similarity and text semantic similarity into one unified hardness score.
- Added FP32 upcast + log-sum-exp for BFloat16 training stability.
- Best model (α=5, β=5): +21.8% relative R@10 improvement; MedR 47→33.

Future directions:

- Scale to full Flickr30K + MS-COCO training sets (GPU cluster).
- Explore multimodal linguistic prior (image + caption jointly).
- Learnable α and β rather than fixed hyperparameters.
- Apply to cross-lingual retrieval where semantic overlap is harder to detect.

Core message: Equal weighting of embedding + linguistic similarity (α=5, β=5) prevents semantically equivalent captions from being treated as hard negatives — 21.8% relative R@10 improvement and MedR 47→33, with no extra learnable parameters.